

Monday Lecture was Midterm.

Maximum Modulus Principle

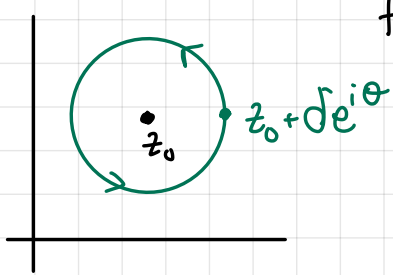
If $f(z)$ is analytic then we know that the modulus is a continuous function of x, y . Hence on any closed region R , $|f(z)|$ has a maximum and that maximum is achieved at some point on the domain.

But actually more is true. In fact, the maximum is always achieved on the boundary and never in the interior.

Lemma: Suppose that $|f(z)| \leq |f(z_0)|$ for each point z in a neighbourhood of z_0 where f is analytic.

Then $f(z) = f(z_0)$ in that neighbourhood.

Proof: Let C be the circle of radius δ around z_0 .
Then


$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_C \frac{f(z)}{z - z_0} dz \\ &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + \delta e^{i\theta})}{\delta e^{i\theta}} \cdot i \delta e^{i\theta} d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \delta e^{i\theta}) d\theta \end{aligned}$$

$$\Rightarrow |f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \delta e^{i\theta})| d\theta \leq |f(z_0)| \quad \text{by assump.}$$

But then
$$\frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + \delta e^{i\theta})| d\theta = |f(z_0)|$$

$$\Rightarrow \frac{1}{2\pi} \int_0^{2\pi} \underbrace{|f(z_0)| - |f(z_0 + \delta e^{i\theta})|}_{\geq 0} d\theta = 0$$

So we find : $|f(z_0)| = |f(z_0 + \delta e^{i\theta})|$ for all $\theta \in [0, 2\pi]$.

Since this is true for all δ small enough, we find equality for all z close to z_0 .

But then by uniqueness, $f(z) = f(z_0)$ for all z in the neighbourhood.

□

Remark: The final claim that, if f is analytic on a domain D

$$|f(z)| = c \text{ a constant for all } z \in D \iff f(z) = \text{constant for all } z \in D.$$

can be proved either by using the Cauchy-Riemann Equations, or by showing that $\overline{f(z)}$ is analytic.

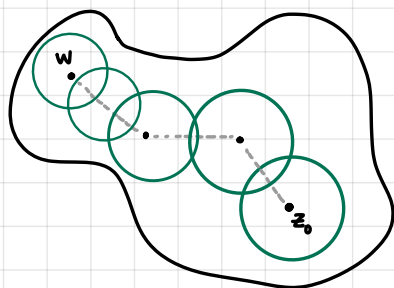
I leave this as an exercise.

The following theorem now tells us how about the location of maxima of analytic functions.

Thm: (Maximum Modulus)

If f is analytic and not constant on a domain D , then $|f(z)|$ has no maximum in D .

Proof: The proof is by contrapositive. We show that if there is a maximum in D , then f is constant.



Let z_0 be the maximum value of f on D . Suppose w is any other point in D .

We want to show that $f(w) = f(z_0)$.

Since D is a domain, we can connect z_0 and w by a polygonal line. Moreover, we can also find a finite sequence of open balls that cover the full line.

By using the Lemma above, we know $f(z) = f(z_0)$ for all z in the first ball. In particular, this means we can find a point z_1 inside the second ball which also has $f(z_1) = f(z_0)$ which is a maximum value of the modulus.

$\Rightarrow$ the Lemma tells us that $f(z) = f(z_0)$ for all z in the second ball. Repeating this argument for all balls we see that $f(w) = f(z_0)$ as claimed.

□

There is some physical intuition for why this is true.

We know that $f(z) = u(x,y) + iv(x,y)$ is a sum of Harmonic functions.

ie. solutions of Laplace's equation.

But Laplace's equation is the time stationary heat equation meaning u, v describe equilibrium distributions of heat in the domain D . We should not expect maximal values inside the domain as the heat equation always tries to equally distribute heat.

On the other hand, we do expect maximal values on the boundary as this is like having a heat source feeding heat into the domain.

Moreover, the function $m(x,y) = \log |f(z)|$ is also Harmonic. So we expect the same behaviour for the modulus.

Question: Is the same true for the minimum values of the modulus?

Taylor Series and Taylor's Theorem

Since we know that every analytic function is infinitely differentiable on its domain, we might suspect it has a Taylor expansion. Indeed this is true.

Thm: (Taylor)

Suppose f is analytic throughout a disc centered at z_0 with radius R_0 . Then f has a power series expansion

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{n!} \frac{d^n f}{dz^n}(z_0)$$

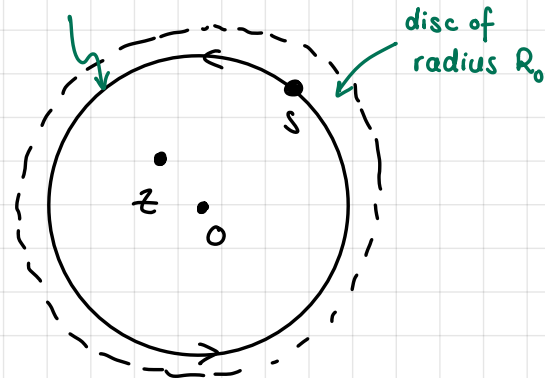
Note in particular, the series converges for all z in the full open disc.

Proof: Without loss of generality assume $z_0 = 0$.

By the Cauchy Integral Formula we know that for any point z inside the circle of radius $r_0 < R_0$ we have:

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(s)}{s-z} ds$$

C : circle radius $r_0 < R_0$



$$\text{But } \frac{1}{s-z} = \frac{1}{s} \frac{1}{1-z/s}$$

$$= \frac{1}{s} \cdot \sum_{n=0}^{\infty} \left(\frac{z}{s}\right)^n \quad \text{which converges for all } \left|\frac{z}{s}\right| < 1 \text{ or } |z| < |s|$$

which is true by assumption

$$= \frac{1}{s} \left(\sum_{n=0}^N \left(\frac{z}{s}\right)^n + \sum_{n=N+1}^{\infty} \left(\frac{z}{s}\right)^n \right)$$

$$= \frac{1}{s} \left(\sum_{n=0}^N \left(\frac{z}{s}\right)^n + \frac{(z/s)^{N+1}}{1-z/s} \right)$$

$$\Rightarrow f(z) = \frac{1}{2\pi i} \int_C \frac{f(s)}{s} \left(\sum_{n=0}^N \frac{z^n}{s^n} + \frac{z^{N+1}}{s^{N+1}} \frac{1}{1-z/s} \right) ds$$

$$= \sum_{n=0}^N z^n \cdot \frac{1}{2\pi i} \int_C \frac{f(s)}{s^{n+1}} ds + z^{N+1} \cdot \frac{1}{2\pi i} \int_C \frac{f(s)}{s^{N+2}(1-z/s)} ds$$

$$= \sum_{n=0}^N \frac{1}{n!} f^{(n)}(0) z^n + \rho_N(z).$$

So if we can show that $\lim_{N \rightarrow \infty} \rho_N(z) = 0$ then we are done.

If we let $|z| = r$ then $|s - z| \geq |s| - |z| = r_0 - r$

$$\Rightarrow \left| \frac{z^{N+1} f(s)}{s^{N+1} (s-z)} \right| \leq \frac{r^{N+1} M}{r_0^{N+1} (r_0 - r)} = \frac{M}{r_0 - r} \left(\frac{r}{r_0} \right)^{N+1}$$

$$\Rightarrow |\rho_N(z)| \leq \frac{2\pi r_0}{2\pi} \cdot \frac{M}{r_0 - r} \cdot \left(\frac{r}{r_0} \right)^{N+1} \\ = \frac{M r_0}{r_0 - r} \left(\frac{r}{r_0} \right)^{N+1} \longrightarrow 0 \quad \text{as } N \rightarrow \infty \\ \text{since } \frac{r}{r_0} < 1.$$

□

Remarks: (1) When a function is entire we can take R to be ∞ and we get convergence on the entire plane.

(2) When a function has a singularity, we can only take R as large as the minimum distance between z_0 and the closest singularity.

(3) We also get an integral formula for the truncation error:

$$\rho_N(z) = z^{N+1} \cdot \frac{1}{2\pi i} \int_C \frac{f(s)}{s^{N+2} (1 - z/s)} ds$$

Example: (a) $e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}$

(b) $\cos z = \frac{1}{2} (e^{iz} + e^{-iz}) = \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} z^{2n}$ (agrees with the real function)

Laurent Series

A natural question then arises: what happens around a singularity?

This depends on the type of singularity. Most singularities occur due to a blow-up of the function. In these cases we can find singular series representations.

Such a series may not be defined at the singular point but is well-defined in a neighbourhood around it.

Examples: (a) Using $e^z = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$ we have:

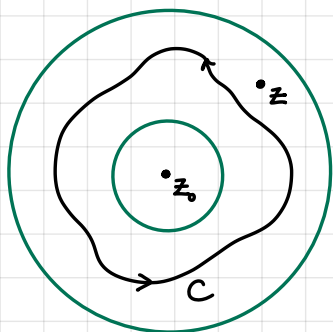
$$f(z) = e^{1/z} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n} = 1 + \frac{1}{z} + \frac{1}{2z^2} + \dots$$

(b) $f(z) = \frac{1}{(z-i)^2}$ is already singular as a series expanded around $z=i$.

$$\begin{aligned} \text{(c)} \quad f(z) &= \frac{-1}{(z-1)(z-2)} = \frac{1}{z-1} - \frac{1}{z-2} = \frac{1}{z-1} + \frac{1}{1-(z-1)} \\ &= \frac{1}{z-1} + \sum_{n=0}^{\infty} (z-1)^n \quad \text{when } |z-1| < 1. \end{aligned}$$

These series are called Laurent series and they always exist and converge in a neighbourhood of an isolated singularity.

Thm: Suppose that a function f is analytic throughout an annular domain $R_1 < |z-z_0| < R_2$ centered at z_0 and let C be a positively oriented simple closed contour around z_0 lying in the annulus. Then, at each point in the domain, $f(z)$ has a series representation:



$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=1}^{\infty} \frac{b_n}{(z-z_0)^n}$$

$$\text{for } R_1 < |z-z_0| < R_2$$

where

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(z)}{(z-z_0)^{n+1}} dz$$

$$b_n = \frac{1}{2\pi i} \int_C f(z) (z-z_0)^{n-1} dz.$$

Note this series expansion is valid for all analytic, even those without singularities. Indeed, if f is analytic interior to C then $f(z)(z-z_0)^{n-1}$ is analytic for all $n \geq 1$ and $b_n = 0$ for all n .

Also note that z_0 itself might not be a singular point, but instead there are singularities contained inside the circle of radius R_1 .

Examples: We can readily verify the above expressions.

(b) $f(z) = \frac{1}{(z-i)^2}$, let C be the curve $z = i + Re^{i\theta}$ $\theta \in [0, 2\pi]$.

$$b_1 = \frac{1}{2\pi i} \int_C \frac{1}{(z-i)^2} dz = 0$$

$$b_2 = \frac{1}{2\pi i} \int_C \frac{1}{(z-i)^2} (z-i) dz = \frac{1}{2\pi i} \int_C \frac{1}{z-i} dz = 1$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{1}{(z-i)^2} (z-i)^{n-1} dz = \frac{1}{2\pi i} \int_C (z-i)^{n-2} dz = 0, \quad n \geq 2.$$

Also we have already shown: $\int_C \frac{1}{(z-i)^{n+2}} dz = 0$ for $n=0,1,2,\dots$

$\Rightarrow f(z) = \frac{1}{(z-i)^2}$ is the full Laurent series around $z=i$.

But we could also find a Laurent series expanded around $z=0$ for example. I leave this as a Homework exercise.

(c) Similarly if $f(z) = \frac{-1}{(z-1)(z-2)} = \frac{1}{z-1} - \frac{1}{z-2}$

and C is the contour $z(\theta) = 1 + \frac{1}{2}e^{i\theta}$ $\theta \in [0, 2\pi]$
we get

$$b_1 = \frac{1}{2\pi i} \int_C \frac{1}{z-1} - \frac{1}{z-2} dz = \frac{1}{2\pi i} (2\pi i + 0) = 1$$

$$b_n = \frac{1}{2\pi i} \int_C \left(\frac{1}{z-1} - \frac{1}{z-2} \right) (z-1)^{n-1} dz = 0 \quad \text{by analyticity for } n \geq 2.$$

$$a_n = \frac{1}{2\pi i} \int_C \left(\frac{1}{z-1} - \frac{1}{z-2} \right) \cdot \frac{1}{(z-1)^{n+1}} dz$$

$$= \frac{1}{2\pi i} \int_C \frac{1}{(z-1)^{n+2}} - \frac{1}{(z-2)(z-1)^{n+1}} dz$$

$$= 0 - \frac{1}{2\pi i} \int_C \frac{1}{(z-2)(z-1)^{n+1}} = - \frac{1}{n!} \frac{d^n}{dz^n} \left(\frac{1}{z-2} \right) \Big|_{z=1}$$

$$= - (-1)^n \frac{n!}{n!} \frac{1}{(z-2)^{n+1}} \Big|_{z=1}$$

$$= - (-1)^n \cdot \frac{1}{(-1)^{n+1}} = 1.$$